

AKANKSHA SHRIVASTAVA

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TECHNICAL SKILLS

Programming & Querying: Python, SQL

Product & Business Analytics Tools: Excel / Google Sheets (advanced), SQL, Tableau, Mixpanel (working knowledge), A/B Testing & KPI Tracking

Machine Learning: TensorFlow, Scikit-learn, Model Training, Evaluation, Explainable AI

Data Engineering & Tools: GitHub, Streamlit, VS Code

Core CS Foundations: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems

Data Science & Analysis: Exploratory Data Analysis, Feature Engineering, ETL Techniques

EXPERIENCE

Indian Institute of Technology Bombay — Summer Fellowship

- Developed and enhanced tool manager for eSIM-related workflows.
- Worked on data organization, tracking, and management for project-related tools and resources.
- Contributed to building and improving the system using Python.

PROJECTS

GridSense AI — Adaptive EV Charging Intelligence Layer — *Python, Data Forecasting, Explainable AI, Streamlit* | [GitHub](#)

- Designed an end-to-end advisor-layer decision-support system that forecasts EV charging demand by zone and hour, applying a closed-loop Predict → Optimize → Recommend → Explain → Improve pipeline for grid planners.
- Built demand-forecasting and peak-detection models with rule-based, feature-importance-driven explainability, generating actionable outputs — demand forecasts, peak alerts, incentive-aware charging recommendations, and infrastructure-siting scores — exported as structured CSV/JSON datasets.
- Developed an interactive Streamlit analytics dashboard with filters, a station-siting map, and pipeline controls, enabling planners to explore zone-level insights and export PDF/PNG snapshots.
- Engineered the pipeline around synthetic, masked data with a decision-support-only design, demonstrating responsible, privacy-conscious data handling for a real-world infrastructure-planning use case.

Live demo: gridsenseai-efyrcrhmvhbeahpmbwstos.streamlit.app

Semantic Search & Fuzzy Clustering System — *Python, Sentence Transformers, FAISS, Fuzzy C-Means, FastAPI*

- Built a semantic search engine over the 20 Newsgroups corpus using 384-dim sentence embeddings (SentenceTransformers/MiniLM) indexed in a FAISS vector database for sub-5ms similarity search, with a documented data-cleaning pipeline to reduce semantic noise.
- Implemented fuzzy c-means clustering for soft, multi-topic cluster assignments; empirically justified cluster count ($k=12$) using the Fuzzy Partition Coefficient and cross-category semantic coherence analysis.
- Designed a cluster-entropy metric to auto-flag boundary-case documents spanning multiple topics, exposed via dedicated analysis and API endpoints.
- Shipped the system as a FastAPI service with a semantic query cache (cosine similarity ≥ 0.85), delivering a 10x speedup on similar queries, plus category-filtered retrieval and cluster-analytics endpoints.

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML) | Vellore Institute of Technology

Sep 2023 - Expected May 2027 | CGPA: 8.97/10

Class XII - 93.4% (2022) | Class X - 95.2% (2020), H.P.D. S.B.V.M., Mathura, UP

CERTIFICATIONS & EXPERIENCE

- Microsoft Certified: SQL AI Developer Associate (July 2026)
- Google – Data Analytics Certification
- IIT Bombay Summer Fellowship, FOSSEE eSim Project (May – July 2026)
- Open Source Contributor – GirlScript Summer of Code (GSSoC) 2026
- AICTE / Shell India / Edunet – AI & Data Analytics (Skills4Future): applied AI/ML to real-world sustainability problems
- Coursera – Applied Machine Learning